
Cello: A Universal Cell-wise Feature Aggregation framework for Reliable Pathology Images Analysis

Hengrui Lou^{1,2} Weihan Li^{1,2,3} Jiazhen Yang^{1,2} Lingxiang Jia^{1,2} Shengxuming Zhang^{1,2,3} Linyun Zhou^{1,2}
Xiuming Zhang⁴ Zhenyang Wang^{2,5} Mingli Song^{1,2} Zunlei Feng^{3,*}

Abstract

Computational pathology has made progress in diagnosis and prognosis prediction from whole slide images (WSIs), yet pipelines still rely on patch-level feature extraction and aggregation, departing from the cell-centric reasoning used by pathologists. This gap limits sensitivity to microlesions and subtle changes, and current methods rarely provide a unified solution that supports both local and global tasks with trustworthy evidence. We propose Cello, a universal cell-wise feature aggregation framework for reliable pathology image analysis. Cello integrates cell-level representations into WSI modeling via protein-signal-supervised cell-wise learning, preserving fine-grained cellular cues under gigapixel constraints. For local tasks, Cello introduces a flexible prototype-based contrastive module for scalable, task-adaptive representation learning. For global tasks, Cello adopts a weakly supervised gated aggregation that can widely leverage WSI labels. Finally, a cell-local-global decision-route consistency objective dynamically aggregates cellular evidence and aligns local predictions with global outcomes, improving reliability and faithfulness. Trained with only hundreds to thousands of samples, Cello achieves performance gains of 3.0% ~ 7.6% and outperforms SOTA pathology foundation models pretrained on tens of thousands of samples. Code is available at <https://github.com/HengruiLou/Cello>.

¹State Key Laboratory of Blockchain and Data Security, Zhejiang University ²Hangzhou High-Tech Zone (Binjiang) Institute of Blockchain and Data Security ³School of Software Technology, Zhejiang University ⁴The First Affiliated Hospital, College of Medicine, Zhejiang University ⁵N-Dimension Tech Co., Ltd.. Correspondence to: Zunlei Feng <zunleifeng@zju.edu.cn>.

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1. Introduction

WSIs provide rich histopathological evidence at gigapixel scale and have become a foundational substrate for computational pathology. Recent advances in weakly supervised learning have enabled accurate slide-level diagnosis and prognosis prediction using only slide-level labels, substantially reducing the burden of dense annotation and facilitating clinical-scale deployment (Ding et al., 2025; Yao et al., 2026). In parallel, large-scale pretraining and pathology foundation models have increasingly accelerated the use of transferable “general-purpose” representations across various tasks and cohorts (Wang et al., 2024; Xu et al., 2025).

A persistent bottleneck, however, arises from the tension between scale and practicality. WSIs cannot be processed end-to-end at native resolution, most pipelines adopt a patch-first strategy: tiling a WSI into patches, extracting patch embeddings, and aggregating them into a slide representation (Zhang et al., 2026). While effective and practical, this patch-level abstraction can obscure cell-centric diagnostic cues. Subtle nuclear atypia, sparse tumor budding, and micro-metastatic foci may occupy only a tiny fraction of a slide yet decisively influence diagnosis; such evidence can be diluted when it is summarized solely through patch embeddings. Graph-based WSI representations have been proposed to bridge local structures and global reasoning (Xiong et al., 2025). Nevertheless, cell/nucleus-centric pipelines commonly require additional supervision (e.g., nuclear masks or cell categories) for robust training (Brussee et al., 2025), and scaling node/edge modeling to gigapixel scenes can be computationally prohibitive. This leaves **Q1: How can cell-level evidence be incorporated into scalable WSI modeling without sacrificing practicality?**

Beyond representation, clinical workflows demand heterogeneous outputs. Different cancer types and reporting protocols require different targets, ranging from local tasks (segmentation, lesion-pattern recognition, tissue composition quantification) to global tasks (cancer diagnosis, subtype classification, outcome prediction). Existing pipelines are optimized for a single granularity: fully supervised local models do not naturally extend to slide-level weak supervision, whereas MIL-style WSI classifiers (Ilse et al., 2018)

are primarily designed for global prediction and provide limited support for plug-and-play local objectives (Aggarwal et al., 2025). This fragmentation raises *Q2: How to build a unified system that flexibly adapts to diverse cancers and diagnostic objectives across local and global levels?*

Finally, reliability and evidence quality are essential for clinical adoption. Pathologists require not only predictions but also supporting evidence aligned with the model’s actual decision process. Many weakly supervised WSI methods provide post-hoc visualizations (often attention heatmaps) as “explanations”, yet attention weights are not inherently faithful (Hu et al.; Wu et al., 2024). Consequently, *Q3: How to connect cell-level evidence, local task semantics, and global decisions while ensuring trustworthy explanations.*

Thus, **Cello** is proposed as a universal cell-wise feature aggregation framework for reliable pathology image analysis. Cello follows cell-first route: cell representations are learned under paired H&E–mIF supervision and aggregated into patch summaries through cross-scale token interaction, allowing cell-centric evidence to be preserved under gigapixel constraints. On top of these, Cello introduces an extensible prototype-based local module that supports heterogeneous local semantics across cancer types by expanding the prototype bank without redesigning the backbone. For slide-level prediction, Cello adopts gated-attention MIL to aggregate patch evidence under weak supervision. To improve reliability, a cell–local–global decision-route consistency objective calibrates global aggregation with prototype-derived evidence and aligns local semantics with global outcomes, enhancing micro-lesion sensitivity and evidence faithfulness. Our key contributions are as follows.

- A cell-first paradigm is introduced, where cellular representations are learned and aggregated before slide-level supervision. This design injects cell-centric evidence into gigapixel modeling without requiring nucleus types or pixel-level masks, improving sensitivity to micro-lesions and subtle cellular changes.
- A universal and extensible cell–local–global framework is developed to accommodate heterogeneous diagnostic requirements across cancer types. A scalable prototype bank is used as an expandable interface for local semantics, enabling cancer- and task-specific local objectives (e.g., subtyping, micro-lesion patterns) to be plugged in without redesigning the backbone.
- A reliability-aware decision route is established to connect cell evidence, local semantics, and global predictions. Prototype-derived evidence is used to calibrate slide-level aggregation, and cross-level consistency constraints align model’s attention, semantic matching, and final decisions for faithful interpretation.

- Comprehensive evaluation is conducted across five cancer types and multiple task granularities (cell/local/global), covering classification, subtyping/grading, and localization-related analyses. Cello consistently improves over strong baselines, including pathology foundation models pretrained on massive data, with gains of 3.0% $\sim$ 7.6% across benchmarks.

2. Related Work

Foundation Models in Computational Pathology. Large-scale pretraining has become a primary driver for improving robustness and transferability in computational pathology. Recent pathology foundation models demonstrate strong cross-dataset generalization and downstream performance by scaling both data and model capacity, including a general-purpose self-supervised encoder (UNI) (Chen et al., 2024) and a visual–language pathology foundation model (CONCH) (Lu et al., 2024). Beyond generic transfer, foundation modeling is also increasingly tailored to slide-level clinical endpoints, e.g., via large-scale weakly supervised training for diagnosis and prognosis prediction (Wang et al., 2024). Meanwhile, slide- and report-aligned multimodal assistants further indicate the potential of language supervision for pathology workflows (Ferber et al., 2025). Despite these advances, most pipelines still rely on patch-first processing and slide-level aggregation, which can under-emphasize cell-centric evidence that is critical for subtle findings.

Cell-centric and Structured Modeling of WSIs. A complementary line of work pursues cell- or nucleus-centric modeling to better capture fine-grained morphology. Modern nuclei instance segmentation and classification models (e.g., transformer-based designs) provide strong cell-level parsing capabilities (Hörst et al., 2024), enabling downstream quantification and spatial reasoning. Graph-based WSI methods further model inter-instance relations by constructing structured representations from detected entities or sampled regions (Li et al., 2024; Zheng et al., 2025b). However, these approaches often depend on additional supervision (e.g., nuclear masks, nucleus types, or handcrafted graph construction rules) and can face scalability limitations when pushing node/edge modeling toward gigapixel scenes. This motivates approaches that can inject cell-level cues into scalable WSI learning without requiring heavy cell annotations or incurring prohibitive graph costs.

Weakly Supervised WSI Learning via MIL Aggregation. Weakly supervised MIL remains the most practical paradigm for gigapixel WSIs, typically aggregating patch embeddings into a slide representation with attention-based pooling or related mechanisms (Zheng et al., 2025a). Many recent variants improve the expressivity of the aggregation module and the modeling of long-range context (including transformer- or graph-augmented MIL designs), yet the

fundamental abstraction is still primarily patch-level (Li et al., 2024). In particular, gated attention pooling is widely used to score instances and perform weighted aggregation for slide prediction, but it inherits the patch-first bottleneck when micro-lesions occupy only a small fraction of the slide. Our work is aligned with the practicality of MIL, while explicitly introducing a cell-wise representation pathway that remains computationally feasible at WSI scale.

Trustworthy Evidence and Interpretable Reasoning.

Clinical deployment requires not only accurate predictions but also evidence that is faithful to the model’s decision process. Attention heatmaps are commonly used as post-hoc explanations in weakly supervised WSI analysis, yet recent studies show that attention-based explanations for modern architectures (including vision transformers) are not inherently faithful and can be misleading under perturbation or reparameterization (Wu et al., 2024; Cai et al., 2025). More broadly, saliency-style explanations may fail updated sanity-check protocols and robustness diagnostics (Hedström et al., 2024). To improve interpretability, prototype-based MIL methods introduce explicit concept-like anchors that can yield more inspectable instance-level reasoning (Tai et al., 2025). Nevertheless, existing interpretable MIL designs typically operate at a single granularity (instance-to-slide) and do not explicitly enforce cross-scale agreement among cell evidence, local task semantics, and global slide decisions. This gap motivates our decision-route consistency learning that aligns evidence across the cell–local–global pathway for more trustworthy interpretations.

3. Method

Cello follows a forward modeling route across three levels—cell, local, and global—and then performs cross-level calibration along a cell–local–global decision route. The cell level injects protein-guided cellular cues into H&E features; the local level builds scalable prototypes to represent diverse local semantics; the global level aggregates patch evidence for slide prediction; finally, local–global calibration and cell–local refinement align evidence and improve reliability.

3.1. Cell-wise Feature Learning and Aggregation

Each WSI $\mathbf{X}$ is tiled into K H&E patches $\{\mathbf{X}_k^{HE}\}_{k=1}^K$. Cellpose-SAM (Pachitariu et al., 2025) is applied to each H&E patch to obtain cell instance masks $\mathcal{M}_k = \{\Omega_{k,i}\}_{i=1}^{N_k}$, where $\Omega_{k,i}$ denotes the region of the i -th cell. Each cell region is treated as an ROI. A Transformer-based cell-wise patch encoder $E_{HE}(\cdot)$ extracts cell tokens $\{\mathbf{T}_{k,i}\}_{i=1}^{N_k}$, a patch token set $\mathbf{PE}_k$, and a patch cls token $\mathbf{CLS}_k^{HE}$ obtained by cross-attention aggregation over cell tokens. The cell tokens and cls token are mapped by an MLP layer $MLP(\cdot)$ to a common latent dimension d so that it can be

aligned with the mIF branch:

$$\overline{\mathbf{CLS}}_k^{HE} = MLP(\mathbf{CLS}_k^{HE}) \in \mathbb{R}^d, \quad (1)$$

$$\overline{\mathbf{T}}_{k,i} = MLP(\mathbf{T}_{k,i}) \in \mathbb{R}^d. \quad (2)$$

For paired samples, an aligned multi-channel mIF patch $\mathbf{X}_k^{mIF} \in \mathbb{R}^{H \times W \times P}$ is available at the same location, where P is the number of protein channels. The same cell regions are used to pool a P -dimensional protein vector per cell as follows:

$$\mathbf{s}_{k,i} = \frac{1}{|\Omega_{k,i}|} \sum_{(p,q) \in \Omega_{k,i}} \mathbf{X}_k^{mIF}(p, q, :) \in \mathbb{R}^P, \quad (3)$$

which is then mapped by a Transformer block $E_{mIF}(\cdot)$ into an mIF cell-token sequence $\{\mathbf{C}_{k,i}\}_{i=1}^{N_k}$ and an mIF cls token $\mathbf{CLS}_k^{mIF} \in \mathbb{R}^d$ obtained by the following cross-attention aggregation:

$$(\{\mathbf{C}_{k,i}\}, \mathbf{CLS}_k^{mIF}) = E_{mIF}(\{\mathbf{s}_{k,i}\}_{i=1}^{N_k}). \quad (4)$$

Cross-modal alignment transfers protein-guided signals into H&E representations via two resolutions: cell-token alignment between corresponding pairs $(\overline{\mathbf{T}}_{k,i}, \mathbf{C}_{k,i})$ and cls-token alignment between patch summaries $(\overline{\mathbf{CLS}}_k^{HE}, \mathbf{CLS}_k^{mIF})$. Alignment is enforced with a bidirectional contrastive objective to encourage mutual adaptation between modalities. Let $\mathcal{B}$ denote paired cell correspondences. For a normalized query $\mathbf{a}$, its positive $\mathbf{b}$, and a reference set $\mathcal{S}$ (tokens of the same type in the current alignment set), define the contrastive term

$$\ell(\mathbf{a}, \mathbf{b}; \mathcal{S}) = -\log \frac{\exp(\mathbf{a}^\top \mathbf{b} / \tau)}{\sum_{\mathbf{s} \in \mathcal{S}} \exp(\mathbf{a}^\top \mathbf{s} / \tau)}. \quad (5)$$

The token-level alignment loss sums the two directions:

$$\mathcal{L}_{\text{tok}} = \sum_{(k,i) \in \mathcal{B}} \left[\ell(\overline{\mathbf{T}}_{k,i}, \mathbf{C}_{k,i}; \hat{\mathcal{S}}_T) + \ell(\mathbf{C}_{k,i}, \overline{\mathbf{T}}_{k,i}; \hat{\mathcal{S}}_C) \right], \quad (6)$$

where $\hat{\mathcal{S}}_T$ and $\hat{\mathcal{S}}_C$ denote mIF and H&E cell token set. Cls-level alignment is defined analogously by replacing $(\overline{\mathbf{T}}_{k,i}, \mathbf{C}_{k,i})$ with $(\overline{\mathbf{CLS}}_k^{HE}, \mathbf{CLS}_k^{mIF})$, and is combined with $\mathcal{L}_{\text{tok}}$ using a weight α to form $\mathcal{L}_{\text{cell}}$.

After alignment, patch-level context is injected back into cell tokens within each modality by reverse cross-attention from the cls token to its cell tokens, which propagates patch-level global cues into cell representations. Both $E_{HE}(\cdot)$ and $E_{mIF}(\cdot)$ remain trainable and are optimized jointly under the alignment objective and downstream tasks.

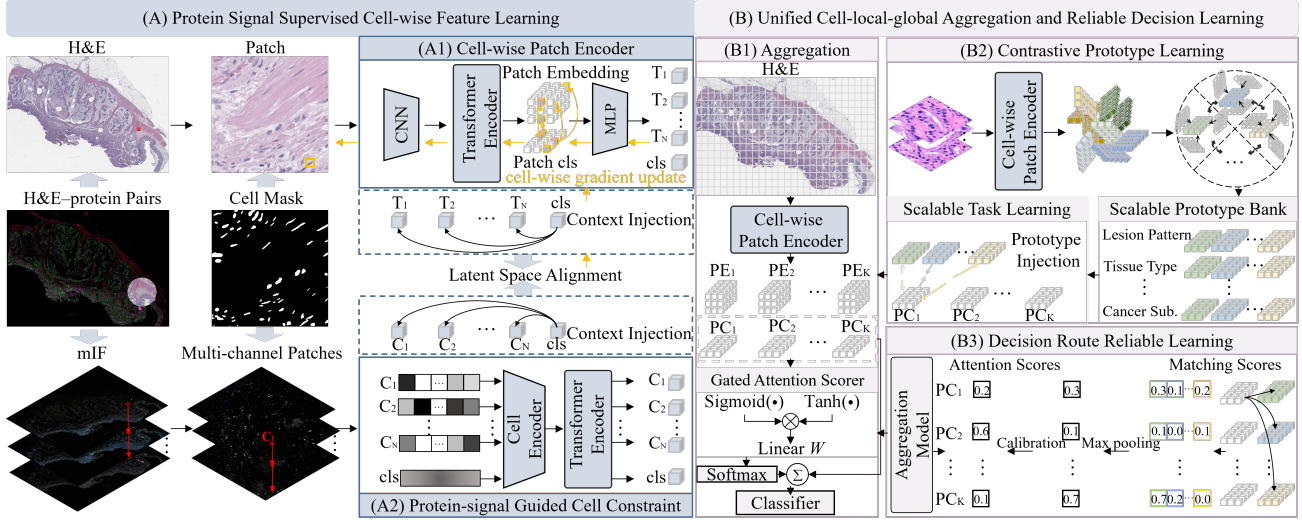


Figure 1. Overview of Cello. (A) With paired H&E–mIF slides, we extract per-cell protein signals from mIF and train a Cell Encoder; its latent alignment supervises the H&E cell-wise Patch Encoder to inject cell-level evidence into patch representations (A1–A2). (B) The cell-aware patch tokens are aggregated for slide-level task using gated-attention MIL (B1), while contrastive prototype learning builds a scalable prototype bank for local tasks and enhances patch discriminability (B2). Decision-route reliable learning calibrates global attention scores via prototype matching and enforces cell–local–global consistency, yielding more faithful and trustworthy evidence (B3).

3.2. Local-level Prototype Contrastive Modeling

Based on the learned cell-wise patch representation $\overline{\text{CLS}}_k^{HE}$, local diagnostic tasks (grading, subtyping, and micro-lesion pattern recognition, et.) can be performed by a classifier. However, conventional classifiers typically operate with a predefined and fixed number of categories, making them inflexible when extending to new lesion types. To address this limitation, we introduce a prototype-based approach in this section, which identifies lesion types by computing the similarity between prototypes and cell-level patch tokens. Specifically, the local-level model represents S lesion types using a scalable prototype bank. Local semantics are carried by the patch cls token $\overline{\text{CLS}}_k^{HE}$. A projection head maps $\overline{\text{CLS}}_k^{HE}$ into a normalized local embedding $\mathbf{v}_k \in \mathbb{R}^d$. K-means is performed on $\{\mathbf{v}_k\}$ with S clusters, and each cluster is assigned to a semantic class by majority vote of local labels. For each class $s \in \{1, \dots, S\}$, dense representatives (closest to centroid) are selected to initialize prototypes $\{\mu_{s,r}\}_{r=1}^R$, where R denotes the total number of prototypes for class s . All prototypes are trainable and normalized as $\hat{\mu}_{s,r} = \mu_{s,r} / \|\mu_{s,r}\|_2$. Training uses a prototype-to-instance contrastive objective: instances are pulled toward prototypes of the same class and pushed away from other classes as follows:

$$\mathcal{L}_{\text{loc}} = - \sum_{k \in \mathcal{D}_{\text{loc}}} \log \frac{\sum_{r=1}^R \exp(\mathbf{v}_k^\top \hat{\mu}_{y_k, r} / \tau_{\text{loc}})}{\sum_{s=1}^S \sum_{r=1}^R \exp(\mathbf{v}_k^\top \hat{\mu}_{s, r} / \tau_{\text{loc}})}, \quad (7)$$

where $\mathcal{D}_{\text{loc}}$ denotes the local cell-wise patch representation set, and τ_{loc} is a hyperparameter. This loss promotes discriminability of $\overline{\text{CLS}}_k^{HE}$ for fine-grained local semantics,

which is critical when micro-lesion evidence is sparse.

3.3. Global-level WSI Modeling

At the global level, K patch classification tokens $\{\overline{\text{CLS}}_k^{HE}\}_{k=1}^K$ are aggregated using gated attention-based multiple instance learning (MIL). A dual-branch gating mechanism (tanh and sigmoid activations) generates an attention feature, which is then projected into an attention logit and normalized via softmax. The resulting slide-level embedding is computed as the attention-weighted sum of the patch tokens, and subsequently fed into a classifier to obtain the whole-slide image (WSI) prediction $\hat{y}_n$, as formulated below:

$$\begin{aligned} \hat{y}_n &= \text{softmax}(\mathbf{W}^{\text{cls}} \mathbf{g}_n + \mathbf{b}^{\text{cls}}), \mathbf{g}_n = \sum_{k=1}^K \omega_k \overline{\text{CLS}}_k^{HE}, \\ \omega_k &= \frac{\exp(\xi_k)}{\sum_{k'=1}^K \exp(\xi_{k'})}, \xi_k = (\mathbf{w}^{\text{att}})^\top \mathbf{a}_k, \\ \mathbf{a}_k &= \tanh(\mathbf{W}^{\text{tanh}} \overline{\text{CLS}}_k^{HE}) \odot \sigma(\mathbf{W}^{\sigma} \overline{\text{CLS}}_k^{HE}). \end{aligned} \quad (8)$$

3.4. Cell–Local–Global Reliable Learning

Reliable learning is organized into two coupled parts. Local–global calibration uses local prototypes to produce explicit semantic evidence for each patch cls token and aligns this evidence with global attention. Cell–local refinement uses the same prototype contrast to back-propagate into the cell-wise encoder through $\overline{\text{CLS}}_k^{HE}$, strengthening micro-lesion sensitivity at the cell aggregation stage.

3.4.1. LOCAL–GLOBAL CALIBRATION

For each patch cls token, similarity to local prototypes yields matching scores over S semantics. Cosine similarity on normalized vectors is used for stability. The matching score of k -th patch to semantic s is defined as the maximum similarity over the R prototypes of that semantic:

$$m_{k,s} = \max_{r \in \{1, \dots, R\}} (\mathbf{v}_k^\top \hat{\boldsymbol{\mu}}_{s,r}). \quad (9)$$

Max-pooling over patches summarizes a slide-level score for each semantic, $v_s = \max_k(m_{k,s})$, which provides an explicit semantic route for interpretation. Meanwhile, global calibration needs a per-patch evidence score, defined as the strongest semantic support in that patch:

$$e_k = \max_{s \in \{1, \dots, S\}} (m_{k,s}). \quad (10)$$

Evidence is then used to calibrate global attention. Specifically, attention logits are shifted by e_k with strength $\rho \geq 0$, producing calibrated attention weights. To align attention with explicit evidence, an evidence distribution is formed by softmax over $\{e_k\}$, and a KL divergence is minimized:

$$\bar{\xi}_k = \xi_k + \rho e_k, \quad \bar{\omega}_k = \frac{\exp(\bar{\xi}_k)}{\sum_{k'=1}^K \exp(\bar{\xi}_{k'})}, \quad (11)$$

$$\mathcal{L}_{\text{lg}} = \text{KL}(\text{softmax}_k(e_k) \parallel \bar{\omega}_n). \quad (12)$$

3.4.2. CELL–LOCAL REFINEMENT

Cell–local refinement injects local semantic discriminability back into the cell-wise patch encoder E_{HE} . At the local level, prototype contrastive learning is applied on the patch cls token $\overline{\text{CLS}}_k^{HE}$ through the projected embedding $\mathbf{v}_k = \text{norm}(q(\overline{\text{CLS}}_k^{HE}))$. Since $\overline{\text{CLS}}_k^{HE}$ is obtained by aggregating the cell-token sequence $\{\mathbf{T}_{k,i}\}$ inside E_{HE} , the local contrastive objective provides a direct training signal that flows from local semantics to cell aggregation. Concretely, for each locally annotated patch $k \in \mathcal{D}_{\text{loc}}$, the gradient of the local loss updates the patch cls token and, through it, the internal parameters of E_{HE} :

$$\theta \leftarrow \theta - \eta \frac{\partial \mathcal{L}_{\text{loc}}}{\partial \theta}, \quad (13)$$

$$\frac{\partial \mathcal{L}_{\text{loc}}}{\partial \theta} = \sum_{k \in \mathcal{D}_{\text{loc}}} \frac{\partial \mathcal{L}_{\text{loc}}}{\partial \overline{\text{CLS}}_k^{HE}} \cdot \frac{\partial \overline{\text{CLS}}_k^{HE}}{\partial \theta}. \quad (14)$$

Here θ denotes the parameters of E_{HE} and η is the learning rate. Because $\overline{\text{CLS}}_k^{HE}$ is a function of both the patch token set $\mathbf{PE}_k$ and the cell tokens $\{\mathbf{T}_{k,i}\}$ (via cross-attention aggregation), the same refinement signal also propagates to the patch tokens and cell tokens inside E_{HE} . As a result,

the encoder is encouraged to produce patch cls representations that are separable under local semantics (including micro-lesion patterns), which in turn sharpens the cell-level evidence preserved during aggregation and improves sensitivity to subtle findings. In addition, stronger separability of $\overline{\text{CLS}}_k^{HE}$ improves prototype matching quality, thereby reinforcing the local–global calibration component.

4. Experiment

4.1. Experiment Setup

Benchmark Datasets. Experiments are conducted on seven WSI cohorts covering five cancer types, including three public benchmarks and four private clinical cohorts. The public datasets include TCGA-CRC (551 WSIs) (Chang, 2013), PANDA (10,616 WSIs) (Bulten et al., 2022), and CPTAC-CCRCC (222 WSIs) (Clark et al., 2019), which provide standardized, widely-used evaluation settings and enable reproducible comparisons with prior work. To further assess clinical robustness under micro-lesion and subtle-morphology scenarios that are often under-represented in public benchmarks, four private cohorts are additionally used: 4500PCas (4,543 WSIs), Prostate104 (104 WSIs), Liver52 (52 WSIs), and Stomach40 (40 WSIs).

Paired H&E–mIF Data. To enable protein-signal guided cell-wise learning, we use five paired H&E–mIF datasets spanning multiple cancer types and multiplex panels. For each case, H&E and mIF images are acquired from the same tissue section, providing intrinsic tissue correspondence without inter-section misalignment. Specifically, we include Rectum19mIF, Kidney7mIF, Prostate9mIF, Liver9mIF, and Gastric15mIF. All mIF images are acquired using a multi-spectral imaging system, and same-section pairing reduces the need for complex cross-section registration.

Evaluation Metrics. We evaluate both diagnostic accuracy and lesion localization. For slide-level classification, we report Accuracy (ACC) and the Area Under the ROC Curve (AUC). For cancer-region localization, we formulate the task as a binary segmentation problem (cancer vs. non-cancer) and report region-level metrics with cancer as the positive class: Dice, Intersection over Union (IoU), Recall (Sensitivity), and Precision (Positive Predictive Value).

Baselines. To align with our modeling scope, we compare against three families of methods: (i) GNN-based models, including Tea-Graph (Lee et al., 2022), Patch-GCN (Chen et al., 2021), and TransGNN (Tang et al., 2024); (ii) MIL-based aggregators, including ZoomMIL (Thandackal et al., 2022), CLAM (Lu et al., 2021), TransMIL (Shao et al., 2021), and ABMIL (Ilse et al., 2018); (iii) Pathology foundation models (PFMs), including CTrans (Wang et al., 2022), HIPT (Chen et al., 2022), UNI (Chen et al., 2024), CHIEF (Wang et al., 2024), and CONCH (Lu et al., 2024).

Table 1. Performance comparison on PANDA, 4500PCas, CPTAC-CCRCC and TCGA-CRC. We report AUC for slide-level diagnosis and grading. **Bold** indicates the best and underline indicates the second best in each column. [†] marks the best result among non-PFM (Pathology Foundation Model) baselines (GNN-based and MIL-based).

Architecture	Methods	Pretraining	PANDA		4500PCas		CPTAC-CCRCC		TCGA-CRC	
		Samples	Diagnosis	Grading	Diagnosis	Grading	Diagnosis	Grading	Diagnosis	Grading
GNN-based	Tea-Graph	–	0.868 [†]	0.836 [†]	0.822	0.740	0.747 [†]	0.723 [†]	0.860 [†]	0.853 [†]
	Patch-GCN	–	0.824	0.818	0.756	0.776 [†]	0.726	0.689	0.823	0.818
	TransGNN	–	0.848	0.817	0.817	0.751	0.612	0.588	0.844	0.838
MIL-based	CLAM	–	0.836	0.816	0.727	0.644	0.579	0.440	0.844	0.838
	ABMIL	–	0.847	0.794	0.729	0.706	0.604	0.578	0.829	0.806
	TransMIL	–	0.847	0.834	0.776	0.734	0.706	0.681	0.829	0.824
	ZoomMIL	–	0.860	0.833	0.834 [†]	0.766	0.646	0.622	0.838	0.831
PFM-based	HIPT	10K+	0.858	0.825	0.790	0.754	0.733	0.697	0.858	0.849
	CTrans	35K+	0.930	0.927	0.819	0.772	0.772	0.742	0.845	0.838
	UNI	100,000K+	0.948	<u>0.944</u>	0.868	0.831	0.824	0.779	0.870	0.866
	CHIEF	60K+	<u>0.953</u>	0.941	<u>0.922</u>	0.844	0.839	0.801	<u>0.896</u>	<u>0.878</u>
	CONCH	117,000K+	0.949	0.934	0.908	<u>0.849</u>	<u>0.842</u>	<u>0.813</u>	0.873	0.862
Ours	Cello	–	0.978	0.950	0.941	0.900	0.857	0.832	0.937	0.916
	Δ vs PFM	✓	+2.6%	+0.6%	+2.1%	+6.0%	+1.8%	+2.3%	+4.6%	+4.3%
	Δ vs non-PFM	✗	+12.7%	+13.6%	+12.8%	+16.0%	+14.7%	+15.1%	+9.0%	+7.4%

4.2. Result Analysis

Experiments are conducted across three levels of tasks: global, local, and cell-level. Global tasks include slide-level cancer diagnosis and cancer grading. Local tasks focus on lesion grading and segmentation. Cell-level evaluation targets the accurate localization of micro-cancerous regions.

4.2.1. GLOBAL CANCER DIAGNOSIS AND GRADING

Table 1 summarizes slide-level AUC for cancer diagnosis and grading on four WSI benchmarks. Across all eight settings, Cello attains the highest AUC. We observe a consistent trend that MIL- and GNN-based pipelines are generally weaker than PFM backbones, highlighting the importance of strong patch representations; however, gains tend to plateau when prediction relies solely on patch-level aggregation. Cello further improves upon the strongest PFM baselines, with larger margins on grading tasks that depend on subtle morphological cues and are thus more sensitive to the loss of cell-level evidence under patch abstraction (+6.0 AUC on 4500PCas grading and +4.6/+4.3 on TCGA-CRC diagnosis/grading), while still yielding competitive gains on datasets closer to saturation such as CPTAC-CCRCC.

4.2.2. LOCAL LESION GRADING AND SEGMENTATION

Table 2 evaluates lesion-region localization by treating model heatmaps as binary cancer masks. Across both PANDA and 4500PCas, Cello yields the strongest region-level performance on all metrics (Dice/IoU/Recall/Precision), indicating that its high-

Table 2. Cancer-region performance on PANDA and 4500PCas. **Bold** and underline denote the best and second best.

Data	Method	Dice	IoU	Recall	Precision
PANDA	HIPT	0.694	0.531	0.744	0.650
	CTrans	0.738	0.584	0.788	0.694
	UNI	<u>0.848</u>	<u>0.736</u>	<u>0.898</u>	<u>0.803</u>
	CHIEF	0.836	0.718	0.886	0.791
	CONCH	0.797	0.662	0.847	0.753
	Cello	0.894	0.808	0.929	0.846
4500PCas	HIPT	0.632	0.461	0.726	0.638
	CTrans	0.703	0.542	0.743	0.665
	UNI	<u>0.816</u>	<u>0.689</u>	<u>0.875</u>	<u>0.784</u>
	CHIEF	0.805	0.673	0.849	0.767
	CONCH	0.801	0.668	0.822	0.721
	Cello	0.837	0.721	0.919	0.852

attention regions align more consistently with true cancer areas. The qualitative comparisons in Fig. 4 corroborate this trend: across WSIs dominated by different Gleason patterns, Cello produces compact and lesion-focused heatmaps, whereas competing methods exhibit either diffuse responses or incomplete coverage. We further assess whether localized regions support fine-grained lesion discrimination via patch-level Gleason pattern classification on 224×224 ROIs (Fig. 2). Using the same frozen-encoder protocol for PFMs, Cello attains the highest average accuracy and

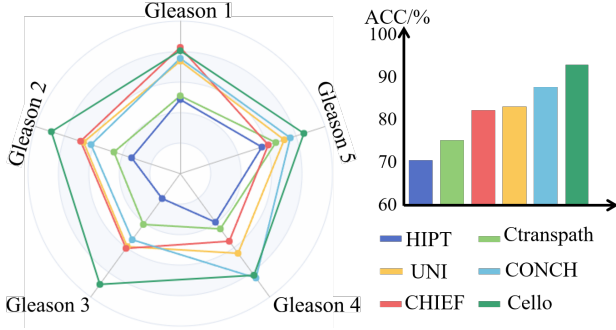


Figure 2. Patch-level Gleason pattern classification accuracy on PANDA (5-way classification; 224×224 patches).

Table 3. Micro-cancer localization under heatmap budgets. We report *Hit@Top-p%* (%) on micro-lesion instances by selecting the top- $p\%$ highest-score patches per WSI. **Bold** and underline indicate the best and second best in each row.

Cancer	Top- $p\%$	UNI	CHIEF	CONCH	Cello
Prostate104	10%	45.2	49.0	<u>54.8</u>	83.7
	30%	<u>59.6</u>	51.9	57.7	91.3
	50%	<u>85.6</u>	74.0	57.7	97.1
Liver52	10%	34.6	28.8	25.0	55.8
	30%	<u>65.4</u>	44.2	30.8	80.8
	50%	<u>76.9</u>	55.8	44.2	100.0
Stomach40	10%	17.5	<u>27.5</u>	12.5	72.5
	30%	<u>42.5</u>	32.5	25.0	87.5
	50%	57.5	55.0	<u>62.5</u>	90.0

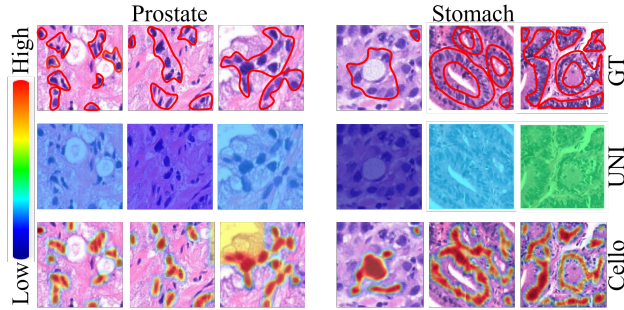


Figure 3. Qualitative micro-cancer localization. Examples from prostate and stomach WSIs comparing Cello with UNI. UNI yields patch-level heatmaps with a single score per patch, while Cello produces cell-resolved evidence maps that capture intra-patch heterogeneity and better align with micro-lesion masks.

shows particularly strong improvements on intermediate patterns (e.g., Gleason 2/3), which are often confusable due to subtle morphological differences.

4.2.3. CELL MICRO-CANCER DETECTION

To directly assess micro-cancer localization, we adopt a budgeted “top- $p\%$ ” setting that emulates limited inspection

Table 4. Ablation study on CPTAC-CCRCC for global WSI diagnosis and grading (AUC). s denotes prototypes’ numbers per class.

Module	Ablation	Diagnosis	Grading
A1	w/o cls align.	0.838	0.801
	w/o tok. align.	0.844	0.829
	w/o dim. map	0.850	0.825
	w/o A1	0.834	0.799
B2	$s = 10$	0.847	0.827
	$s = 5$	0.855	0.826
	$s = 1$	0.853	0.830
	w/o B2	0.840	0.801
B3	w/o local-global	0.842	0.815
	w/o cell-local	0.852	0.824
	w/o B3	0.842	0.812
FULL	Cello	0.857	0.832

capacity. **Cello uses a stricter hit rule:** a micro-lesion is counted as hit only if (i) at least 50% of a connected lesion component is covered by selected patches, *and* (ii) the predicted heatmap mask overlaps the ground-truth lesion mask by $\geq 25\%$; baselines require coverage only. For each WSI, we rank patches by their heatmap scores and select the top- $p\%$ patches as predicted suspicious regions ($p \in \{10, 30, 50\}$). A micro-lesion instance is counted as *hit* if at least 50% of its connected component is covered by the selected patches. To avoid overstating gains from finer-grained outputs, we impose a *stricter* criterion on Cello: in addition to coverage, the predicted heatmap mask must overlap the ground-truth lesion mask by at least 25%.

Table 3 summarizes Hit@Top- $p\%$ across three cancers. Despite the stricter hit definition, Cello consistently achieves substantially higher hit rates under all budgets. The advantage is most pronounced in the low-budget regime (Top-10%), where micro-lesions are easiest to miss under patch-only abstraction: for example, on stomach WSIs Cello reaches 72.5% Hit@10%, whereas the best baseline remains below 30%. As the budget increases, all methods improve, but Cello maintains a clear margin and approaches saturation on liver (100% at Top-50%), indicating that its evidence calibration helps prioritize truly informative regions rather than dispersing attention over visually confounding tissue.

Fig. 3 provides representative examples illustrating *why* Cello improves micro-lesion detection. PFM-based baselines yield patch-level heatmaps that cannot resolve intra-patch heterogeneity, since each patch is assigned a single score. Cello, by contrast, produces cell-resolved evidence within high-scoring patches and can selectively emphasize a small number of highly indicative cells. This cell-aware evidence provides a more precise localization signal for micro-lesions and aligns with our goal of faithful, fine-grained diagnostic support under gigapixel constraints.

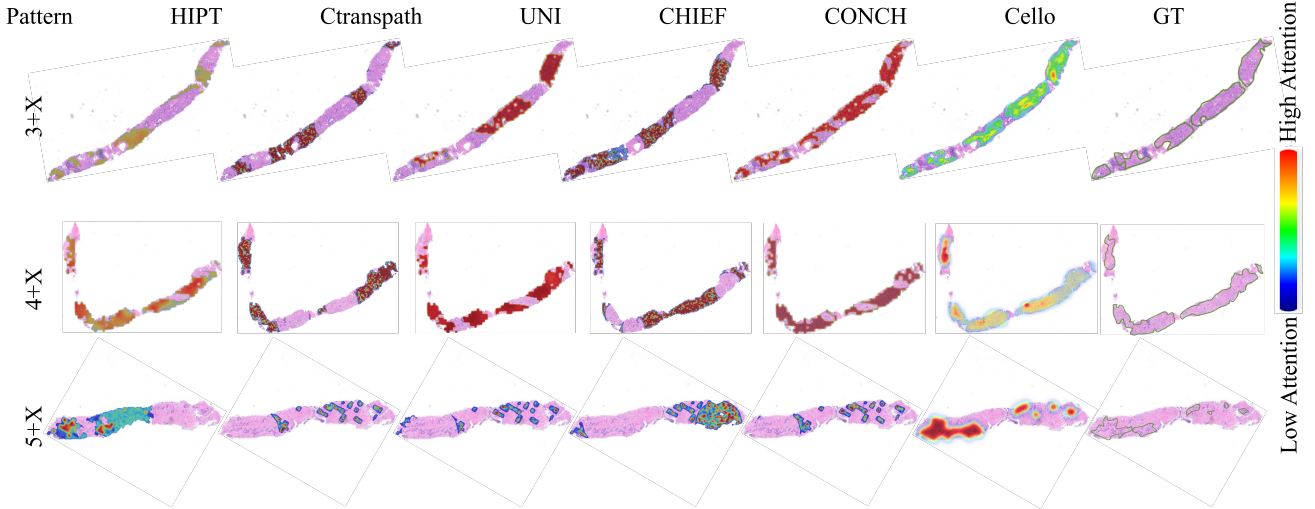


Figure 4. Qualitative lesion localization heatmaps. Heatmap comparison between Cello and PFM-based methods on prostate WSIs dominated by Gleason 3+X, Gleason 4+X, and Gleason 5+X, illustrating their lesion localization behavior across different patterns.

4.3. Ablation Study

To validate the unified cell–local–global pipeline, global WSI prediction is treated as the end-to-end target and key components that inject cell-level evidence and local semantics into the global decision are ablated. Table 4 reports results on CPTAC–CCRCC; ablations on TCGA–CRC and the corresponding analysis are provided in Appendix A.

A1: Cell-wise Feature Learning and Aross-Modal Alignment. Removing the full A1 module (patch-only) leads to consistent drops (diagnosis/grading: $0.857 \rightarrow 0.834$ / $0.832 \rightarrow 0.799$), confirming cell-wise aggregation provides complementary gains beyond patch-only encoding. Among A1 components, cls-level alignment is more influential than token-level (grading: $0.832 \rightarrow 0.801$ vs. $0.832 \rightarrow 0.829$), suggesting that aligning patch-level summaries stabilizes how cellular evidence is aggregated for slide-level prediction. Removing the dimension mapping also degrades performance (grading: $0.832 \rightarrow 0.825$), indicating that a shared latent space is important for cross-modal supervision.

B2: Local Prototype Contrast and Extensible Semantics. Disabling B2 results in a clear degradation (diagnosis/grading: $0.857 \rightarrow 0.840$ / $0.832 \rightarrow 0.801$), showing that explicit local semantics regularization benefits global endpoints even under weak supervision. Varying the number of prototypes per class indicates that multiple prototypes are helpful but not overly sensitive on this dataset: $s \in \{5, 10\}$ remains close to the full model, while $s = 1$ slightly reduces representational capacity and yields a modest drop.

B3: Decision-route Reliable Learning. Removing B3 decreases performance (grading: $0.832 \rightarrow 0.812$), supporting the role of decision-route calibration in stabilizing attention-based aggregation. Separating two routes

shows that local–global calibration drives most of the gain (grading: $0.832 \rightarrow 0.815$), while cell–local refinement provides a smaller but consistent improvement (grading: $0.832 \rightarrow 0.824$), consistent with its role in back-propagating local discriminability into the cell-wise encoder.

5. Discussion and Future Work

Despite progress, challenges remain. Pathology supervision is often noisy due to subjective grading criteria and mixed-pattern lesions. A practical extension is to make the prototype module (B2) noise-aware by introducing dedicated noise prototypes and a difficulty-aware memory that separates easy from hard samples. Additionally, Cello’s current “cell-level” reasoning is implemented at the token level within detected cell regions. A promising direction is to move toward instance-adaptive cell representations (e.g., contour-aware embeddings), so evidence maps better conform to cellular geometry while retaining scalability.

6. Conclusion

In this paper, we presented Cello, a universal framework that bridges the critical granularity gap in computational pathology by integrating cell-level reasoning into scalable WSI modeling. By leveraging protein-signal supervision from paired mIF data, Cello successfully injects molecularly grounded cellular cues into H&E representations without incurring the computational cost of gigapixel graph construction. Furthermore, our novel integration of task-scalable prototype learning and a rigorous cell–local–global decision-route consistency objective fundamentally addresses the reliability bottleneck of conventional MIL paradigms. Extensive experiments across multiple cancer types demon-

strate that Cello not only achieves state-of-the-art diagnostic performance—particularly in detecting subtle microlesions—but also provides faithfully calibrated evidence that aligns with clinical intuition. We believe Cello offers a principled path toward more transparent, precise, and trustworthy AI assistants for clinical pathology workflow.

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Impact Statement

This work proposes a novel computational pathology framework to enhance the interpretability and precision of whole-slide image analysis. All experiments were conducted using either public datasets or private data obtained with institutional review board approval and in compliance with relevant ethical guidelines. By improving the detection of subtle lesions and providing more transparent reasoning, this research has the potential to support pathologists and reduce diagnostic variability. While the direct societal consequences are anticipated to be beneficial—such as aiding in standardizing diagnostic workflows—we acknowledge that the integration of AI tools into clinical practice requires careful validation, stakeholder engagement, and ongoing assessment of real-world impact.

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Cello: A Universal Cell-wise Feature Aggregation Framework for Reliable Pathology Image Analysis

—Supplementary Material—

In the supplementary material, we provide comprehensive additional details, including the full results omitted from the main-text metric figures (A), detailed experimental settings and hyperparameter configurations (B), an in-depth description of the datasets (C), and visualizations from real-world deployment of *Cello* (D).

A. Detailed Experiments Results

Table 5. Detailed results corresponding to Fig. 2 in the main text under *Local lesion grading and segmentation* (5-way classification on PANDA using 224×224 patches). Best result in each row is highlighted in bold.

Class	HIPT	CTransPath	UNI	CHIEF	CONCH	Cello
Gleason 1	0.743	0.755	0.869	0.914	0.878	0.903
Gleason 2	0.668	0.729	0.834	0.844	0.808	0.945
Gleason 3	0.601	0.706	0.796	0.803	0.768	0.949
Gleason 4	0.697	0.724	0.823	0.774	0.923	0.912
Gleason 5	0.782	0.830	0.859	0.804	0.880	0.927
Mean	0.698	0.749	0.836	0.828	0.851	0.927

Analysis. As the detailed breakdown for Fig. 2, Table 5 shows that *Cello* attains the best mean accuracy (0.927), exceeding the strongest baseline on average (CONCH, 0.851) by **+7.6** percentage points. The improvements are largest on Gleason 2 and Gleason 3, where *Cello* surpasses the best competing method (CHIEF) by **+10.1** and **+14.6** points, respectively, indicating markedly stronger recognition of these more challenging mid-grade patterns. *Cello* also achieves the top accuracy on Gleason 5, improving over CONCH by **+4.7** points. Meanwhile, *Cello* is marginally below the best method on Gleason 1 and Gleason 4 (both by 1.1 points), suggesting that its gains primarily come from harder categories rather than classes that are already well separated. Overall, *Cello* provides a strong and consistent advantage in patch-level Gleason pattern classification, driven by substantial gains on Gleason 2–3 and competitive performance across all classes.

Ablation Analysis on TCGA-CRC. Table 6 provides additional ablation results on TCGA-CRC.

A1: cell-wise feature learning and cross-modal alignment. Removing A1 leads to clear drops in both diagnosis and grading (w/o A1: $0.937 \rightarrow 0.902$ / $0.916 \rightarrow 0.881$), confirming that cell-wise aggregation benefits global prediction. Among A1 components, cls-level alignment contributes more than token-level alignment (grading: $0.916 \rightarrow 0.883$ vs. $0.916 \rightarrow 0.899$), indicating that patch-level summary alignment is critical for stabilizing aggregation under weak supervision. Removing the dimension mapping $MLP(\cdot)$ also reduces grading performance ($0.916 \rightarrow 0.909$), supporting the need for a shared latent space for cross-modal supervision.

B2: local prototype contrast. B2 is particularly important on this dataset: disabling B2 causes a large drop (diagnosis/grading: $0.937 \rightarrow 0.858$ / $0.916 \rightarrow 0.827$), suggesting that global endpoints depend strongly on localized and fine-grained semantics. The number of prototypes per class shows a clear capacity effect: $s \in \{5, 10\}$ remains close to the full model, while $s = 1$ sharply degrades both diagnosis and grading, consistent with heterogeneous local patterns requiring multiple prototypes to cover multi-modal semantics.

B3: decision-route reliable learning. Removing B3 leads to a pronounced drop (w/o B3: $0.937 \rightarrow 0.878$ / $0.916 \rightarrow 0.870$), highlighting the importance of decision-route calibration. Local–global calibration accounts for most of the benefit (w/o local–global: $0.937 \rightarrow 0.910$ / $0.916 \rightarrow 0.883$), while cell–local refinement yields a smaller but consistent gain (w/o cell–local grading: $0.916 \rightarrow 0.899$), matching its role of propagating local discriminability back into the cell-wise encoder.

B. Implementation Details

For protein-signal supervised cell-wise learning, we instantiate the patch-level and cell-level backbones with CTrans and ViT, respectively, and obtain cell masks using Cellpose-SAM. For WSI aggregation, we build the gated-attention module on

Table 6. Additional ablation study on TCGA-CRC for global WSI diagnosis and grading (AUC). A1: cell-wise feature learning and cross-modal alignment; B2: local prototype contrast; B3: decision-route reliable learning. In B2, s denotes the number of prototypes per semantic class.

Module	Ablation	Diagnosis	Grading
A1	w/o cls align.	0.905	0.883
	w/o tok. align.	0.913	0.899
	w/o dim. map (m)	0.915	0.909
	w/o A1 (patch-only)	0.902	0.881
B2	$s = 10$	0.935	0.911
	$s = 5$	0.933	0.914
	$s = 1$	0.869	0.853
	w/o B2	0.858	0.827
B3	w/o local-global	0.910	0.883
	w/o cell-local	0.930	0.899
	w/o B3	0.878	0.870
FULL	Cello	0.937	0.916

top of the ABMIL-style architecture. All methods use identical train/validation/test splits with a ratio of 7:1:2; PFMs and ImageNet-pretrained CNN baselines are fine-tuned under the same protocol. Unless otherwise stated, we train all models with Adam for 50 epochs using a learning rate of 10^{-4} and cross-entropy loss. We adopt early stopping with a patience of 20 epochs (training stops if the validation performance does not improve for 20 consecutive epochs) to mitigate overfitting. For localization-related tasks, the attention decision threshold is set to 0.5.

Inference experiments were conducted on a computing cluster consisting of eight servers, each equipped with NVIDIA GeForce RTX 5090 GPUs (32 GB memory) and dual AMD EPYC 9654 CPUs (96 physical cores per processor), running Ubuntu 22.04.5 LTS. Training is conducted on 8xA800 GPUs, with a batch size of 4 per GPU. Data augmentation includes random horizontal flipping and keep-ratio random resizing. ALL experiments were implemented using PyTorch 2.9.0 as the deep learning framework.

Alignment: 10 epochs with about 1.72M paired HE-mIF patches per cancer on average. For latent-space alignment in Eq. (6), we use symmetric InfoNCE-style bidirectional contrastive loss at two resolutions: cell-token and cls-token alignment. In each patch, we sample up to 256 valid cell tokens to form the reference set; positives are matched HE/mIF cell tokens, while negatives are other tokens of the same type within the current alignment set.

Cross-attention transformer: Patch-level cls token is obtained by cross-attention aggregation over cell tokens, and the reverse cross-attention injects patch-level context back into cell representations. HE aggregator is a lightweight Transformer block with depth = 2, 8 attention heads, and FFN dimension = 2048. The projection layer $MLP(\cdot)$ in Eqs. (1)–(2) is a 2-layer MLP with GELU, mapping features into the shared latent dimension $d=1280$.

Prototype module: We project $\overline{\text{CLS}}_k^{HE}$ into normalized local embeddings $\mathbf{v}_k$, perform K-means with S clusters, assign each cluster to a semantic class by majority vote of local labels, and initialize the trainable prototypes $\mu_{s,r}$ using dense representatives nearest to each centroid.

mIF Channels: Cello does not force into universal channel space. We process it with cohort-specific $E_{mIF}(\cdot)$ branch before mapping into shared latent space. Marker panels differ across cohorts, and forcing cross-cohort channel sharing introduces negative transfer. We verified this in controlled PANDA experiments: prostate-specific supervision performed best, while cross-cancer channel sharing consistently degraded performance. Lesion-labels: CRC and CCRCC are used for global-level. Data used for B2 are PANDA and 4500PCas. For PANDA, lesion labels comes from officially provided annotation masks for Gleason patterns. 4500PCas come from private clinical used in clinically registered deployment.

C. Dataset Details

Paired H&E-mIF data. To enable protein-guided cell-wise learning, paired H&E-mIF data are further collected across multiple cancers and multiplex panels. For each case, the H&E and mIF images are acquired from the *same tissue section*

Table 7. Summary of paired H&E–mIF datasets used for protein-guided cell-wise learning.

Cancer type / Cohort	#Pairs	#Channels	Channel names
Rectum19mIF	40	19	Hoechst, AF1, CD31, CD45, CD68, Argo550, CD4, FOXP3, CD8a, CD45RO, CD20, PD-L1, CD3e, CD163, E-cadherin, PD-1, Ki67, Pan-CK, SMA
Kidney7mIF	90	7	DAPI, CD20, CD23, CD3, FOXP3, CD8, PAX8
Prostate9mIF	6	9	CKpan, P53, Ki-67, P504s, PSA, ERG, AR, P63, S100
Gastric15mIF	6	15	CD138, MPO, D2-40, vimentin, CD34, CKpan, P53, Ki-67, HER2, CEA, CDX2, MUC5AC, MUC6, CLDN18.2, Syn
Liver9mIF	6	9	P53, Ki-67, HepPar-1, Arginase-1, AFP, CK7, CK19, GPC3, HBS

(i.e., the same slide), providing intrinsic tissue correspondence without inter-section misalignment and reducing the need for complex cross-section registration. The paired datasets are summarized as follows. (i) **Rectum19mIF** contains 40 paired slides with a 19-plex panel (Hoechst, AF1, CD31, CD45, CD68, Argo550, CD4, FOXP3, CD8a, CD45RO, CD20, PD-L1, CD3e, CD163, E-cadherin, PD-1, Ki67, Pan-CK, SMA). (ii) **Kidney7mIF** comprises 90 paired slides with a 7-plex panel (DAPI, CD20, CD23, CD3, FOXP3, CD8, PAX8). (iii) **Prostate9mIF** includes 6 paired slides with a 9-plex panel (CKpan, P53, Ki-67, P504s, PSA, ERG, AR, P63, S100). (iv) **Gastric15mIF** includes 6 paired slides with a 15-plex panel, covering shared cell-type markers—CD138 (plasma cells), MPO (neutrophils), D2-40 (lymphatic endothelium), vimentin (fibroblasts), and CD34 (vascular endothelium)—as well as gastric- and phenotype-specific markers (CKpan, P53, Ki-67, HER2, CEA, CDX2, MUC5AC, MUC6, CLDN18.2, and Syn). (v) **Liver9mIF** includes 6 paired slides with a 9-plex panel (P53, Ki-67, HepPar-1, Arginase-1, AFP, CK7, CK19, GPC3, and HBS). All mIF images are acquired using a multispectral imaging system.

D. Real-world Deployment Examples

Due to confidentiality agreements, we only present WSI-level qualitative results from three real-world *Cello* deployment scenarios that we have been explicitly permitted to disclose. Specifically, Fig. 5 and Fig. 6 illustrate detection results for lung adenocarcinoma and prostate adenocarcinoma, respectively, using a visualization protocol that differs from that in the main text. In real-world deployment, false negatives are typically more critical than false positives; therefore, we visualize attention scores starting from 0 to emphasize low-attention regions that may correspond to missed lesions. In contrast, Fig. 7 corresponds to a secondary AI review of cases initially screened as negative by junior pathologists; here we display only high-confidence regions (attention > 0.5), revealing a missed microcarcinoma.

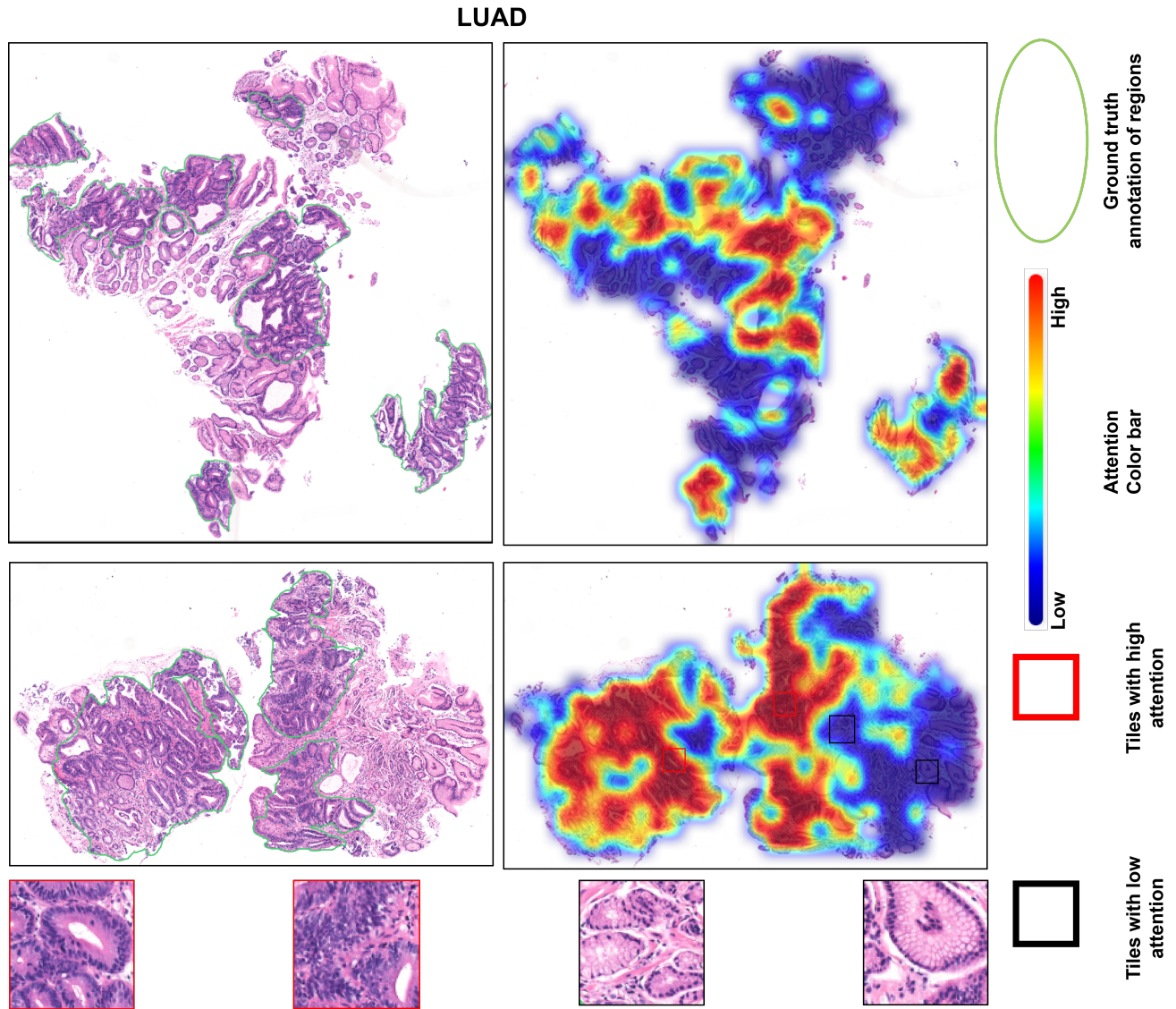


Figure 5. Qualitative lesion localization results of Cello on LUAD. comparing lesion regions annotated by expert pathologists and regions identified by Cello highlights morphologically abnormal areas that show high spatial concordance with pathologist-defined tumor regions, demonstrating accurate lesion localization in lung adenocarcinoma.

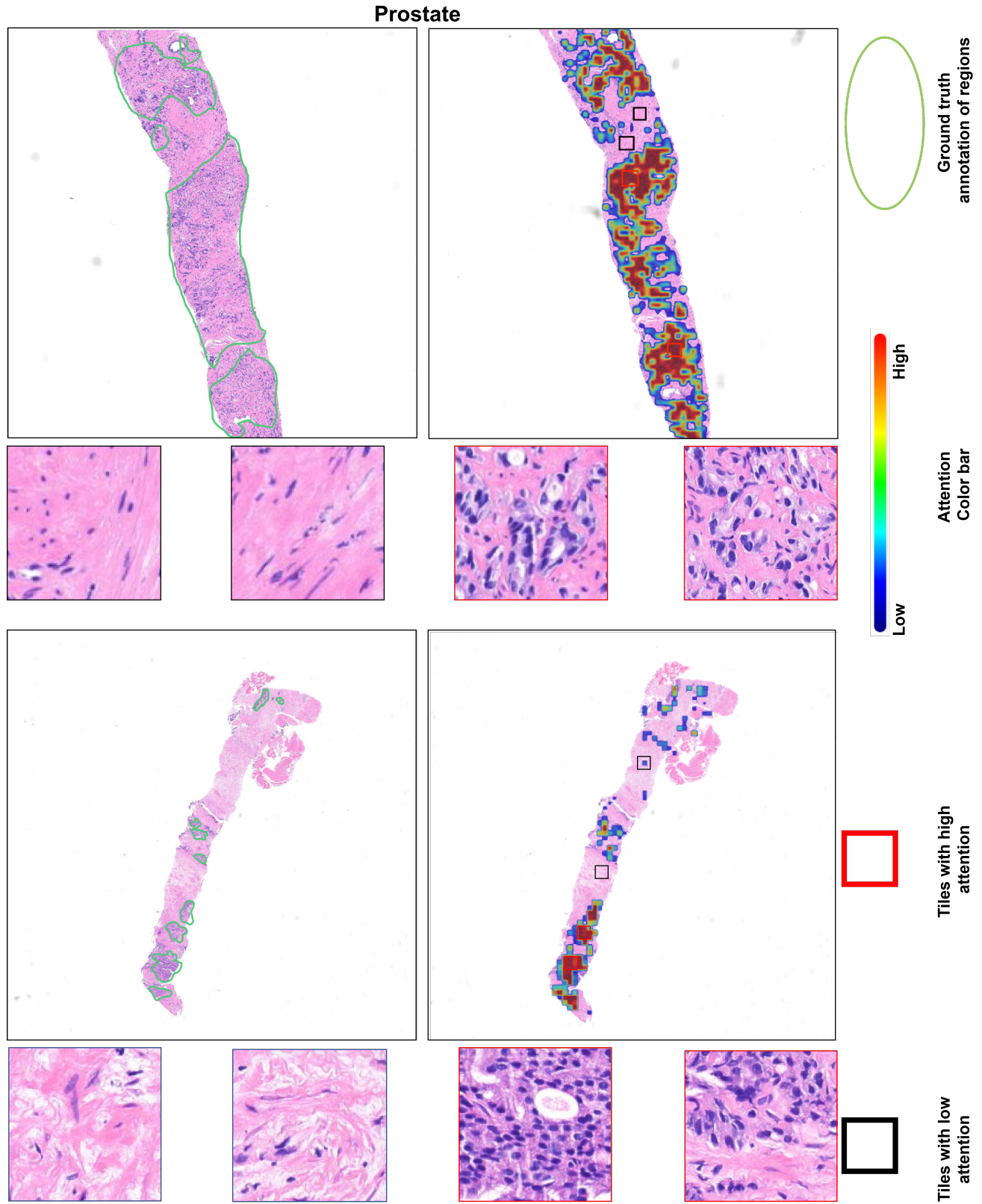


Figure 6. Qualitative lesion localization results of Cello on Prostate. comparing lesion regions annotated by expert pathologists and regions identified by Cello highlights morphologically abnormal areas that show high spatial concordance with pathologist-defined tumor regions, demonstrating accurate lesion localization in prostate cancer.

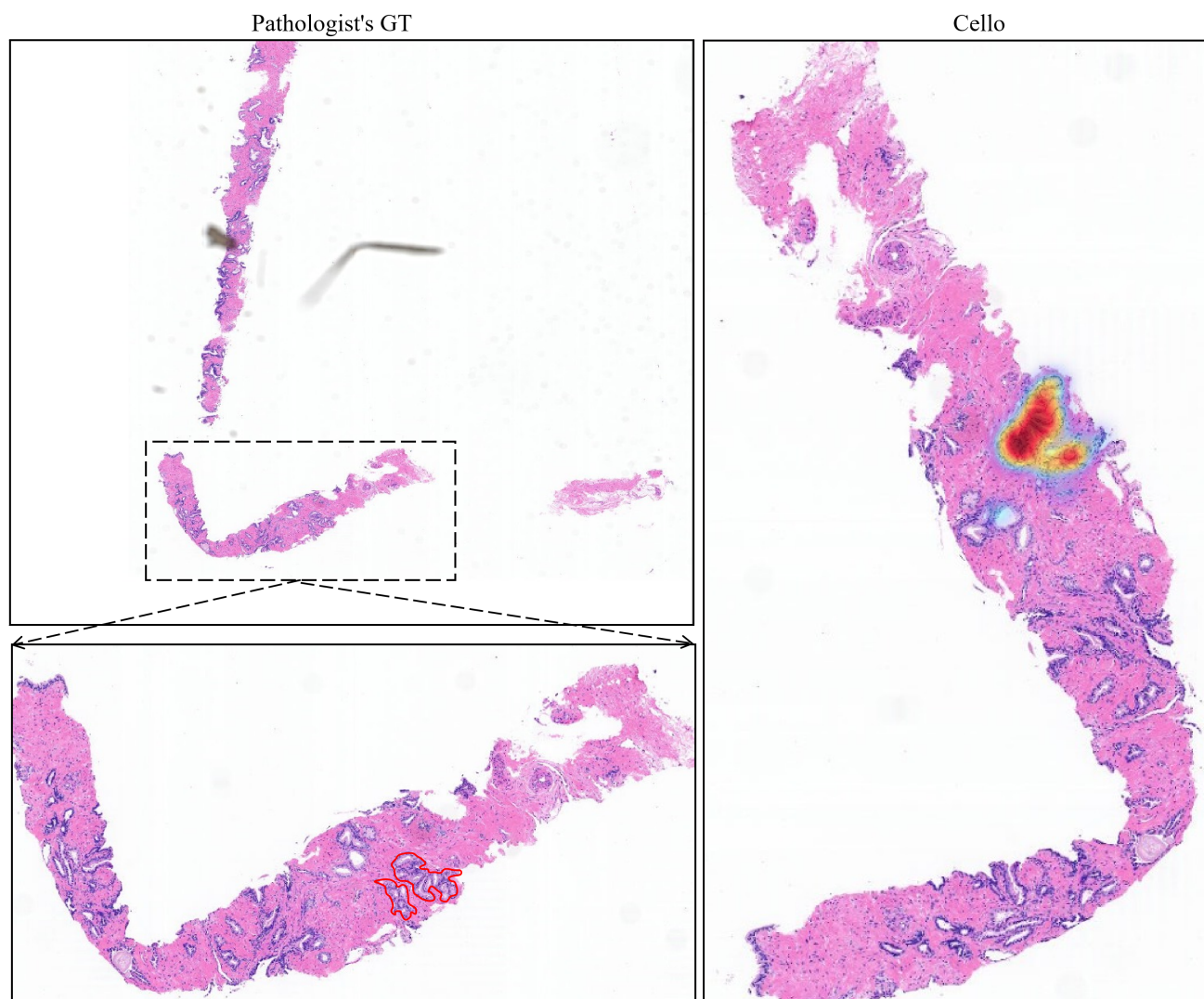


Figure 7. Example of Cello correcting a missed microcarcinoma by a junior pathologist.